

Liouville Equation for an Interacting Classical Gas

Consider a dilute classical gas with Hamiltonian

$$H = \underbrace{\sum_{i=1}^N \left[\frac{p_i^2}{2m} + U(q_i) \right]}_{\equiv H_1(\{p_i, q_i\})} + \frac{1}{2} \sum_{j \neq i} V(q_i - q_j) \equiv H_1(\{p_i, q_i\}) + \frac{1}{2} \sum_{i=1}^N \sum_{j \neq i} V(q_i - q_j)$$

Liouville's theorem tells us that the phase space density ρ is constant along classical trajectories (i.e. total derivative of ρ vanishes):

$$\frac{d}{dt} \rho(\{p_i(t), q_i(t)\}, t) = 0$$

Classical mechanics tells us that the time evolution of any observable is related to the Poisson bracket with the Hamiltonian via:

$$\frac{dO}{dt} = \{O, H\} + \frac{\partial O}{\partial t}$$

(if you haven't seen this before, use the chain rule and Hamilton's EOM.)

(or take classical limit of Heisenberg EOM)

where

$$\{f, g\} \equiv \sum_{i=1}^N \left(\frac{\partial f}{\partial q_i} \cdot \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \cdot \frac{\partial g}{\partial q_i} \right)$$

Thus

$$\frac{d\rho}{dt} = 0 = \{\rho, H\} + \frac{\partial \rho}{\partial t}$$

The Poisson bracket is bilinear, so $\{f, a g_1 + g_2\} = a \{f, g_1\} + \{f, g_2\}$, and thus

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \{\rho, H\} &= -\frac{1}{2} \sum_i \sum_{j \neq i} \{\rho, V(q_i - q_j)\} \\ &= \frac{1}{2} \sum_i \sum_{j \neq i} \sum_k \frac{\partial \rho}{\partial p_k} \frac{\partial V(q_i - q_j)}{\partial q_k} \\ &= \frac{1}{2} \sum_i \sum_{j \neq i} \left(\frac{\partial \rho}{\partial p_i} \frac{\partial V(q_i - q_j)}{\partial q_i} + \frac{\partial \rho}{\partial p_j} \frac{\partial V(q_i - q_j)}{\partial q_j} \right) \end{aligned}$$

Use symmetry of V :

$$\frac{\partial S}{\partial t} + \{S, H_1\} = \sum_{i=1}^N \frac{\partial S}{\partial p_i} \sum_{j \neq i} \frac{\partial V(q_i - q_j)}{\partial q_i} \quad (1)$$

The BBGKY Hierarchy

The s -body probability density is obtained as a marginal over the $(s+1, s+2, \dots, N)$ coordinates

$$S_s(q_1, \dots, q_s; p_1, \dots, p_s) \equiv \int \prod_{k=s+1}^N d^3 p_k d^3 q_k S(\{q_i, p_i\}).$$

This is a probability density — it is convenient to define:

$$f_s(\{p_i, q_i\}) = \frac{N!}{(N-s)!} S_s(\{p_i, q_i\})$$

which is now a number density:

$$\int f_1(q, p) d^3 q d^3 p = 1$$

$$\int f_1(q, p) d^3 q d^3 p = N.$$

We can integrate (1) over $\prod_{i=2}^N d^3 p_i d^3 q_i$ to obtain the dynamics of f_1 :

$$\partial_t f_1(q_1, p_1) + \{f_1, H_1\} = \int d^3 q_2 d^3 p_2 \frac{\partial V(q_1 - q_2)}{\partial q_1} \frac{\partial}{\partial p_1} f_2(q_1, p_1, q_2, p_2) \quad (2)$$

In general, you will find that the dynamics of f_s depends on the dynamics of f_{s+1} . So the exact dynamics is not closed and requires knowledge of f_N :

$$\partial_t f_1 = G_1[f_2]$$

$$\partial_t f_2 = G_2[f_3]$$

$\vdots$

$$\partial_t f_{N-1} = G_{N-1}[f_N]$$

The goal of the Boltzmann equation is to approximate (2) by expressing the RHS in terms of f_1 alone (or some coarse-grained $\hat{f}_1$).

Time scales and Coarse-Graining

The free evolution $\{f_i, H_i\}$ occurs on an intrinsic timescale τ_F . This diverges with system size because it is related to the length scale of variations of $U(q)$. For a box of linear dimension L , $\tau_F \sim \frac{L}{v}$.

Suppose the interaction potential V has a characteristic length $d \ll L$. Collisions relax on a timescale τ_c — this is related to the mean free time. Since τ_c depends only on d and the free volume b/w particles $1/n$, it does not change with system size.

So for large systems, $\tau_F \gg \tau_c \rightarrow$ separation of timescales.

We will coarse grain on a length scale l , $l^3 \sim V_c$
 $d \ll l \ll L$

to generate $\hat{f}_i$:
$$\hat{f}_i = \frac{1}{V_c} \int_{V_c} d^3q f_i(q, p, t)$$

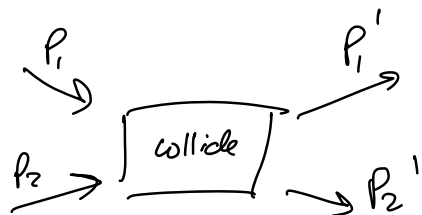
If we apply this averaging to (2), the LHS keeps the same form

$$\partial_t f_i + \{f_i, H_i\} \rightarrow \partial_t \hat{f}_i + \{\hat{f}_i, H_i\}$$

The RHS must be done phenomenologically. Make two assumptions:

1) Dilute gas $\rightarrow$ only binary collisions

2) Particles are uncorrelated before collision, $f_2(q_1, p_1, q_2, p_2) \approx f_1(q_1, p_1) f_1(q_2, p_2)$



$$\begin{aligned} \partial_t \hat{f}_i + \{\hat{f}_i, H_i\} &= \int d^3p_2 \int d^2\sigma |\vec{v}_1 - \vec{v}_2| \left[\hat{f}_i(q_1, p_1') \hat{f}_i(q_1, p_2') - \hat{f}_i(q_1, p_1) \hat{f}_i(q_1, p_2) \right] \\ &= C[\hat{f}] \end{aligned}$$

(3)

The H-theorem

Consider

$$H[F] = \int d\vec{p} d\vec{q} f(\vec{p}, \vec{q}, t) \log f(\vec{p}, \vec{q}, t)$$

Theorem: If f solves the Boltzmann Equation, then $\frac{dH}{dt} \leq 0$.

Proof: See lecture 7. Briefly

$$\frac{d}{dt} H(t) = \int d\vec{p} d\vec{q} \partial_t [f \log f] = \int d\vec{p} d\vec{q} (\log f + 1) \partial_t f$$

$$= \int d\vec{p} d\vec{q} \log f \partial_t f$$

$$= \underbrace{\int d\vec{p} d\vec{q} \log f C[f]}_{\downarrow} - \underbrace{\int d\vec{p} d\vec{q} \log f \{f, H_1\}}_{\downarrow 0}$$

$$= \frac{1}{4} \int d\vec{p}_1 d\vec{p}_2 d\vec{q} d\vec{r} (f_1' f_2' - f_1 f_2) [\log f_1 f_2 - \log f_1' f_2']$$

$$\log \frac{f_1' f_2'}{f_1 f_2} > 0 \text{ iff } f_1' f_2' > f_1 f_2$$

The Steady State

Suppose $H[f]$ has stationarized. Then $\int d\vec{p} d\vec{q} \log f C[f] = 0$

$\Downarrow$

$$\log f_1 f_2 - \log f_1' f_2' = 0$$

$$\log f_1 + \log f_2 = \log f_1' + \log f_2'$$

So $\log f_i$ is an additive collisional invariant. There are 5 such quantities:

→ Particle number

→ 3 components of momentum $\vec{p}_1 + \vec{p}_2 = \vec{p}_1' + \vec{p}_2'$

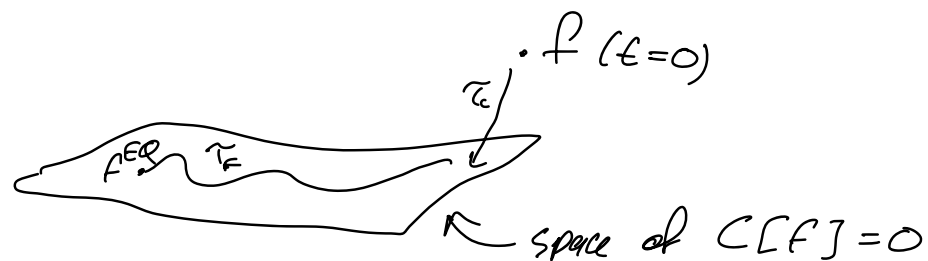
→ Energy $\frac{p^2}{2m} + U(q)$

$$\log f_i(\vec{q}, \vec{p}) = \gamma(\vec{q}) - \vec{\alpha}(\vec{q}) \cdot \vec{p} - \beta(\vec{q}) \cdot \left[\frac{\vec{p}^2}{2m} + U(\vec{q}) \right]$$

$$f_i(\vec{q}, \vec{p}) = N(\vec{q}) e^{-\vec{\alpha}(\vec{q}) \cdot \vec{p} - \beta(\vec{q}) H_i(\vec{q}, \vec{p})} \Rightarrow \text{local eq.}$$

Any f in the above form satisfies $C[f]=0$. Such f are said to be in local equilibrium.

Global Equilibrium is achieved when $\partial_t f=0$. In a time scale τ_c , f relaxes to f^{LEQ} and so $C[f]$ rapidly vanishes. We then have a slow evolution $\sim \tau_F$ until $\{f, H\}=0$, in which case $\partial_t f=0$.



Can show $\{f^{eq}, H\}=0 \Rightarrow f_1^{eq}(q, p) = N_0 e^{-\beta_0 H_1(q, p)}$

Hydrodynamics

From now on, set $U(q)=0$. Then (3) becomes

$$\partial_t f + \frac{\partial f}{\partial q} \cdot \frac{\vec{p}}{m} = \int d\vec{p}_2 d^3\sigma |\vec{v} - \vec{v}_2| (f'f'_2 - ff_2)$$

We can examine the particle density field: $n(\vec{q}) = \int d^3p f(p, q)$

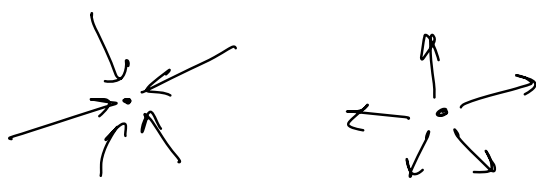
We find (by differentiating and using BE)

$$\partial_t n = -\partial_\alpha (n \vec{u})_\alpha \quad \text{Divergence of a flux} \rightarrow \text{conservation law}$$

$$\text{where } \vec{u}(q) = \frac{1}{n(q)} \cdot \int d\vec{p} \frac{\vec{p}}{m} f(q, p) = \langle \vec{v}(q) \rangle$$

We can define a material derivative $D_t \equiv \partial_t + u_\alpha \partial_\alpha$. In this notation

$$\begin{aligned} D_t n &= -n \partial_\alpha u_\alpha \\ &= -n \nabla \cdot \vec{u} \end{aligned} \rightarrow \text{density changes as div of velocity field.}$$



The velocity evolves as:

$$m D_t u_\alpha = -\frac{1}{n} \partial_\beta P_{\alpha\beta} \quad (5) \quad \text{where} \quad P_{\alpha\beta} = nm \langle (v_\alpha - u_\alpha)(v_\beta - u_\beta) \rangle$$

$$\text{Def } \delta \vec{v} \equiv \vec{v} - \vec{u}$$

$$\text{Then } P_{\alpha\beta} = n \langle \delta v_\alpha \cdot \delta v_\beta \rangle$$

↓
like a covariance matrix
for velocities

Pressure tensor is understood as follows: for a unit vector $\hat{n}$, $P \hat{n}$ gives the momentum flux through a small surface normal to $\hat{n}$.

$P_{\alpha\beta}$ is flux of α component of momentum along β direction.

So 5 is just 3 continuity equations for each component of momentum.

$$m D_t u_\alpha = -\frac{1}{n} \nabla \cdot P_\alpha$$

We can also look at kinetic energy density:

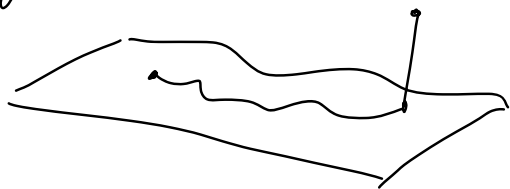
$$\partial_t \varepsilon + u_\alpha \partial_\alpha \varepsilon = -\frac{1}{n} \left[\partial_\alpha h_\alpha + P_{\alpha\beta} u_{\alpha\beta} \right] \quad (6)$$

$$h_\alpha \equiv \frac{nm}{2} \langle \delta v_\beta \delta v_\beta \delta v_\alpha \rangle \quad u_{\alpha\beta} \equiv \frac{1}{2} [\partial_\alpha u_\beta + \partial_\beta u_\alpha]$$

Equations 4, 5, and 6 are simply enforced by the symmetries and conservation laws. The forms of P and h must be obtained from BE.

Zero-Order hydrodynamics

Here we assume $f = f^{LEQ}$, and so $C[f] = 0$ instantaneously. We lose all details of the gas.



$$f^{LEQ} = N(q, t) \exp \left[-\vec{\alpha}(q, t) \cdot \vec{p} - \beta(\vec{q}, t) \frac{\vec{p}^2}{2m} \right]$$

This is Gaussian in p .

$$n(\vec{q}) \equiv \int d^3p f^{LEQ}$$

$$u(q) \equiv \int d^3p \frac{\vec{p}}{m} f^{LEQ} = \langle \vec{v} \rangle / n(q)$$

$$\varepsilon(q) = \frac{1}{2} \frac{m}{n} \int f^{LEQ} \left(\frac{p}{m} - u(q) \right)^2 d^3p = \frac{m}{2} \langle (\vec{v} - \vec{u}(q))^2 \rangle$$

Get $f^{LEQ}(q, p, t) = \frac{n(\vec{q})}{[2\pi mkT]^{3/2}} e^{-\frac{(p-mu)^2}{2mkT}}$ lecture 9

Using this we find: $\varepsilon \equiv \frac{3}{2} kT$

$$P_{\alpha\beta} = nk_B T \delta_{\alpha\beta}$$

Pressure tensor is diagonal $\rightarrow$ only $P_{\alpha\alpha}$ along $\hat{n}$.

$$h_\alpha = 0$$

heat flux zero, temp inhomogeneities do not relax.

Suppose we have a uniform gas, $n(q) = n_0$, $T(q) = T_0$, and we shear it

$$\vec{U}(q) = u(y) \hat{x}$$

We have:

$$\partial_t T + u_\alpha \partial_\alpha T = -\frac{2}{3} T \partial_\alpha u_\alpha$$

$$\partial_t u_\alpha + u_\beta \partial_\beta u_\alpha = -\frac{1}{mn} \partial_\alpha [nkT] = 0 \quad \text{at time } t=0, \text{ since } T=T_0.$$

$$u_\beta \partial_\beta u_\alpha = u(y) \partial_x u_\alpha = 0 \Rightarrow \partial_t u_\alpha = 0 \quad \text{at time } 0.$$

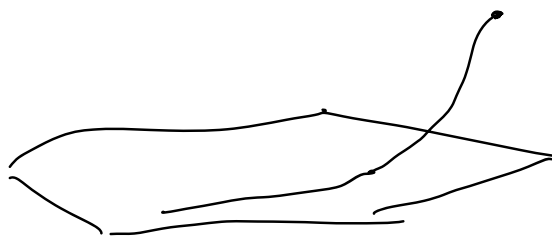
$$\text{Also } \partial_t T = -\frac{2}{3} T \partial_\alpha (u(y) \hat{x}) = -\frac{2}{3} T \partial_x u(y) = 0 \quad \text{at time } 0.$$

First Order Hydro

Define $L(F) = \partial_t F + \{F, H_1\}$.

where have

$$L(F) = C[F] \quad \begin{matrix} \hookrightarrow 1/\tau_F & \hookrightarrow 1/\tau_c \end{matrix}$$



Zeroth order fails when $C \sim \tau_c/\tau_F$.

We will perform a perturbation theory: $F = F_0 + \frac{\tau_c}{\tau_F} F_1 = f_0 + \delta f$.

To make progress, we also approximate

$$C[F] \simeq -\frac{F_1}{\tau_F}$$

$$\begin{aligned}
 & \overset{S_0}{\underbrace{\partial_t f_0 + \{f_0, H_1\}}_{1/\tau_F}} + \underbrace{\frac{\tau_c}{\tau_F} \partial_t f_1 + \frac{\tau_c}{\tau_F} \{f_1, H_1\}}_{\frac{\tau_c}{\tau_F} \cdot \frac{1}{\tau_F}} \overset{\rightarrow \text{diagonal}}{=} - \underbrace{\frac{f_1}{\tau_F}}_{\frac{1}{\tau_c} \cdot \frac{\tau_c}{\tau_F} \sim \frac{1}{\tau_F}}
 \end{aligned}$$

We get corrections to P_{AB} , h_A . Off diagonal terms in pressure tensor give

$$\partial_t u(y) = \frac{\tau_c kT}{m} \frac{\partial^2}{\partial y^2} u(y)$$

for shear modes. $\Rightarrow$ diffusion relation.

This defines viscosity $\mu = n \tau_c kT$.

We can also compute thermal conductivity

$$h_A = -\kappa \partial_{q_A} T$$

with
$$\kappa = \frac{5}{2} \frac{n}{m} \tau_c k^2 T$$